Monsij Biswal

Senior Year, Electronics and Communication Engineeering
National Institute of Technology, Durgapur, IN
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RESEARCH INTERESTS

Wireless Communication, Signal Processing, Neural Networks

EDUCATION

National Institute of Technology, Durgapur, *India*
Bachelor of Technology in Electronics and Communication Engineering — *July 2016 – Present*
GPA: 9.51 / 10.00

Kendriya Vidyalaya, IIT Kharagpur, *India*
High School - Computer Science — *March 2014 – March 2016*
Percentage : 93.80

PUBLICATIONS

- A. U. Rahman, **M. Biswal**, "Error-tolerant Beam Steering of mmWave Antennas by Trajectory Estimation of Highway Vehicles", 11th International Conference on Communication Systems & Networks (COMSNETS), Bengaluru, India, 2019
- A. Sen, **M. Biswal**, S. Datta "Intelligent Traffic Routing Based on Real-time Congestion Analysis", M. V. Chauhan Student Paper Contest, 16th IEEE India Council Conference Conference,Gujarat, 2019 *(in press)*

WORK EXPERIENCE

Christian Albrechts Universität zu Kiel, *Germany*
Research Intern, DAAD WISE Scholar — *May 2019 – July 2019*

Indian Institute of Technology, Kharagpur, *India*
Summer Intern — *May 2018 – June 2018*

PROJECTS

Millimeter Wave Channel Modelling and Performance Indicators
Mentor: Associate Prof. Aniruddha Chandra — *August 2019 – Present*
National Institute of Technology Durgapur
- Channel modelling based on channel sounding measurements in a vehicular & outdoor scenario centered around a specific license free band of 60GHz.
- Extensive analysis of small scale and large scale fading parameters and their effects on rms delay spread of the channel.

Custom Neural Network Training Function for Optical Transmission System
Mentor: Prof. Dr.-Ing. Stephan Pachnicke — *May 2019 – July 2019*
Christian Albrechts Universität zu Kiel
- Analyzed various neural network training algorithms for equalizing a higher order modulation format in a optical transmission system.
- Implemented a custom training function to improve the accuracy of the constellation diagram.

Interactive GUI for ROI Selection of Medical Images
Mentor: Prof. Sudipta Mukhopadhyay *December 2018 – January 2019*
Indian Institute of Technology, Kharagpur

- Framed a GUI in MATLAB for interactive selection of ROI from frontal image samples of mouth cancer victims.
- Classified an image as malignant or benign based on the output of image processing algorithm.

Data Regeneration codes for distributed storage systems
Mentor: Prof. Sant Sharan Pathak *May 2018 – July 2018*
Indian Institute of Technology, Kharagpur

- Implemented and tested various efficient storage algorithms to control data reliability and redundancy.
- Developed regenerating codes, a class of codes to prevent errors during data reconstruction and retrieval in a distributed storage system.

HONOURS AND AWARDS

- **DAAD WISE Scholarship Recipient** *January 2019*
  Was one among 80 students selected for a funded summer internship at Germany
- **MITACS Globalink Research Intern Award** *January 2019*
  Invited to participate in a research project at Laval University, Quebec
- **16th KVS Junior Mathematics Olympiad Awardee** *October 2014*
  For being among the top 30 candidates in India
  Invited to attend training camp for Indian National Mathematics Olympiad (INMO)

SKILLS & OTHERS

Programming: Python, C/C++, Julia
Softwares: MATLAB, CorelDRAW

RELEVANT COURSES

EC-601: Transmission Lines
EC-503: Digital Signal Processing
EE-431: Control Systems Engineering
EC-401: Digital Electronics
CS-331: Programming and Data Structures
EC-501: Digital Communication
EC-504: Computer Architecture and Organization
EC-402: Analog Communication
EC-302: Signals and Systems
MA-331: Engineering Mathematics

POSITIONS OF RESPONSIBILITY

Secretary, IEEE Student Branch, NIT Durgapur *April 2019 – Present*
Being a direct handshake with IEEE, IEEE Student Branch aims to promote research related activities among the youths of the campus. These include events such as seminars, workshops, guest lectures and internship bootcamps.

- Organized short-term course on MATLAB and various workshops for autonomous robotics.
- Collaborated with various other student branches for organising national level events.

Representative, Department of Training and Placement, NIT Durgapur *February 2018 – Present*

- Managed end-to-end recruitment process of various companies visiting the campus including logistics and hospitality.
- Ensured proper exchange of information between students and the business executives.